

University of Oklahoma Computer Science
CS 4233/5233: Software Engineering with AI
Fall 2026

Course Format/Meeting Time and Location

Class delivery: Traditional In-Person

Class meetings: MWF 9 - 9:50 am

Class location: [Physical Science Ctr 0323](#)

Final exam: December 15, 2026, T 8 - 10 am - Physical Science Ctr 0323

Instructor: Abdulhak, Mansoor **Email:** m.hak@ou.edu

TA: Bijaya K C **Email:** Bijaya.K.C-1@ou.edu

Student/Office Hours

MW, 12:00 PM-1:00 PM, DEH 234 or ([Virtually](#))

Student hours are for you. They are times in which you may ask questions about course content, delve deeper into the topic through conversation with me, share concerns you have about the course, and/or explore career or graduate school opportunities with me in this field. I often hear from students that they don't want to interrupt my time in my office, but please know that I want you to come by! In addition to our class time together, these hours not only help you but help me to know the questions my students have so that I may adjust my teaching approach as needed.

Learning Management System/website

This class will use Canvas software for our home page. The URL for the home page is <http://canvas.ou.edu>. Login with your 4+4 using your standard OU password. If you have difficulty logging in, call 325-HELP. This software provides a number of useful features, including a list of assignments and announcements, an electronic mailing list, and grade book. The Canvas course site will be used for all updates. You should check the site regularly.

Course Prerequisite

CS 3203 Software Engineering

Course Description

This course explores how AI can empower software engineers by enhancing productivity, automating repetitive tasks, and promoting architectural and strategic thinking. Students will learn to work effectively with AI tools while maintaining critical thinking and human oversight. The course prepares students to thrive in a rapidly evolving industry by focusing on modern software engineering practices, DevOps, business alignment, and AI tool evaluation.

Course Learning Outcomes (CLOs)

By the end of this course, students will be able to:

1. Apply architectural thinking to design scalable and maintainable systems.

2. Automate workflows using DevOps practices to improve software delivery.
3. Align engineering solutions with business objectives and communicate impact.
4. Use AI-powered tools in development and critically evaluate their output.

ABET Learning Outcomes Alignment

ASO 1	Analyze a complex computing problem and to apply principles of computing and other relevant disciplines to identify solutions.
ASO 3	Communicate effectively in a variety of professional contexts.
ASO 4	Recognize professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.
ASO 6	Apply computer science theory and software development fundamentals to produce computing-based solutions.

Course Learning Outcome (CLO)	Bloom's Level	ABET Student Outcomes (SOs)
Design software architectures with modularity, scalability, and maintainability in mind.	Apply, Analyze, Create	SO1 (Complex problem-solving)
Implement automated development workflows using DevOps practices and modern tools.	Apply, Analyze	SO6 (Experimentation and tools)
Align software solutions with business goals and effectively communicate technical impact.	Analyze, Evaluate, Create	SO3 (Communication), SO4 (Business, ethics)
Use AI-powered tools in development and critically evaluate their output for correctness.	Analyze, Evaluate	SO1 (Problem-solving), SO2 (Design within constraints), SO4 (Ethical responsibility)

Texts and Materials

Introduction

- | | |
|------------|---|
| Required 1 | <ol style="list-style-type: none"> 1. Tenbergen, B., & Krusche, S. (2024). The future of software engineering education and training in the age of AI. IEEE Software, 41(2), 28–29. https://doi.org/10.1109/MS.2023.3345960 2. M. Hamza, D. Siemon, M. A. Akbar and T. Rahman, "Human-AI Collaboration in Software Engineering: Lessons Learned from a Hands-On Workshop," 2024 IEEE/ACM International Workshop on Software-Intensive Business (IWSiB), Lisbon, Portugal, 2024, pp. 7-14, doi: 10.1145/3643690.3648236. 3. Khemka, M., & Houck, B. (2024). Toward effective AI support for developers: A survey of desires and concerns. Queue, 22(3), 60. https://doi.org/10.1145/3675416 |
|------------|---|

- I. Khojah, R., Mohamad, M., Leitner, P., & de Oliveira Neto, F. G. (2024). Beyond code generation: An observational study of ChatGPT usage in software engineering practice. *Proceedings of the ACM on Software Engineering*, 1(FSE), Article 81. <https://doi.org/10.1145/3660788>

Module 1: Architectural Thinking

Esposito, M., Li, X., Moreschini, S., Ahmad, N., Černý, T., Vaidhyanathan, K., ... & Taibi, D. (2025). Generative AI for software architecture: Applications, trends, challenges, and future directions. *arXiv*.
<https://arxiv.org/abs/2503.13310>

Module 2: Business Alignment

Takeuchi, H., & Yamamoto, S. (2020). Business analysis method for constructing business–AI alignment model. *Procedia Computer Science*, 176, 1312–1321. <https://doi.org/10.1016/j.procs.2020.09.140>

Module 3: Continuous DevOps Alignment

Taibi, D., Lavazza, L., & Lenarduzzi, V. (2023). Continuous alignment between software architecture design and development in CI/CD pipelines. In P. Pelliccione, R. Kazman, I. Weber, & A. Liu (Eds.), *Software Architecture* (pp. 61–78). Springer. https://doi.org/10.1007/978-3-031-36847-9_4

Module 4: Critical Review of AI Output

Layman, L., & Vetter, R. (2024). Generative artificial intelligence and the future of software testing. *Computer*, 57(1), 27–32.
<https://doi.org/10.1109/MC.2023.3306998>

Supplementary 2 Krusche, S. (2024). Strategic software engineering: Software engineering beyond the code. Springer. <https://doi.org/10.1007/978-3-031-55670-8>

Supplementary 3 Pressman, R. S., & Maxim, B. R. (2019). *Software Engineering: A Practitioner's Approach*

Teaching Philosophy

My teaching methods include a variety of up-to-date techniques including active participation via an inverted classroom and experiential learning through project-based instruction and assessment. Through these methods, I seek to make courses imitate the work environment as much as possible in order to best prepare students for their careers.

Learning Activities, Assignments, and Assessments

During the semester, students will have:

- **Class activities:** The course schedule includes weekly readings. While you are free to study ahead, following the schedule will ensure you are prepared for in-class activities and homework assignments. These activities are in-class submission.
- **Assignments:** Homework assignments will give you the opportunity to develop essential course skills and assess your understanding of the material. Each assignment instructions, along with its duration, will be posted on the course Canvas page. It will be assessed on the student's submission.
 - This course is highly practical, consisting of 7 assignments split into two phases:
 1. **Preparation:** You will be tasked with preparing a plan and gathering the necessary tools to ensure quality and test your software product.
 2. **Execution:** You will then implement the plan and perform the tests to achieve the desired outcomes.
- **Midterm & Final Exam:** The exam will assess your overall understanding of the course material and the skills you've developed throughout the semester. Detailed information, including the format and schedule, will be posted on the course Canvas page. It will be assessed on the student's submission.

Assessments

Methods	Quantity	Weighting
System Design	1	20%
Workflow Automation with DevOps	1	20%
Business Impact Proposal	1	20%
Midterm Exam	1	10%
Project Presentation	1	10%
Final Exam	1	20%

Assigning Grades

The final letter grading for the course will be as follows: A $\geq$ 90%, B = 75-89%, C = 60-74%, D = 50-59%, F = < 50%. The instructor will round all averages to two significant figures (69.5 will round to 70 and 69.4 will round to 69) to determine the student's letter grade in the course (70 = C, 69 = D). Canvas is equipped with a grade book that preserves the raw data utilized for computing your course grade. It is crucial that you routinely verify the accuracy of your recorded grades. In the event of any identified discrepancies or disagreement, promptly notify me, and I shall promptly address and rectify the matter.

Communication

1. The primary method of communication outside of class will be through a Discord server. The server link will be shared on Canvas. All general questions related to the learning outcomes of the class are encouraged to be discussed openly within the appropriate channels on Discord. However, for questions involving personal matters, participants are

welcome to send private messages within the Discord server for a more confidential interaction.

2. Urgent announcements will be communicated through Canvas. It is your responsibility to regularly check Canvas for updates.
3. For formal communication, please use email to contact me. To facilitate this communication PLEASE, Ensure that you include the semester, the course code ID, the group ID and your Sooner ID (e.g. Fall2026-CS4233-GroupA-123456789 or Fall2026-CS5233-GroupA-123456789) before the subject in your email. Without this information, your message may not be noticed or entertained.

Tentative Schedule

Week	Topic	Reading Chapters	Assessment Method	Total Marks
1	Introduction: Why Software Engineers Still Matter	- The evolving role of the software engineer - What AI automates vs. what humans lead - The new value of system thinking, DevOps mastery, and business fluency - Critical review of AI-generated solutions	- Group discussion: “What AI can’t do” - Reflection: “My future role as a software engineer” - Case study: Bad outcomes from unchecked AI code	
2	Module 1: Architectural Thinking	Designing modular, extensible software	- Redesign a real-world app with different architecture strategies CS5: Conduct an independent literature review on AI tools for software architecture.	
3	Module 1: Architectural Thinking	Architecture trade-offs and documentation	Draw and present architecture diagrams	
4	Module 1: Architectural Thinking	Patterns for scalability and maintainability	- Use any open source AI tool to propose and critique designs - Compare 3 different AI tools	20
5	Module 2: Workflow Automation with DevOps	Git, CI/CD, testing automation	Set up an automated pipeline from	

			scratch with Git Copilot	
6	Module 2: Workflow Automation with DevOps	Containerization and deployment pipelines	Analyze errors and improve pipeline reliability with Git Copilot	
7	Module 2: Workflow Automation with DevOps	Monitoring, rollback, and alerting	- Compare human-scripted CI/CD with AI-assisted configuration - CS5: Analyze different prompt engineering strategies for automating workflows with DevOps tools.	20
8	Module 3: Business-Aligned Engineering	Engineering priorities in business context	Translate a product backlog into business goals	
9	Module 3: Business-Aligned Engineering	Communicating technical value to stakeholders	- Build a pitch deck for a technical feature - CS5: Lead one seminar discussion Business goals, connecting readings to practice.	20
10	Midterm Review Week	- Peer reviews of architectural and pipeline designs - Team presentations: "Our Modern Engineering Stack"	Q&A: "What Industry Now Expects from Software Engineers"	10
11	Module 4: AI-Augmented Software Development	Using AI tools to generate code.	Bias, quality, and ethical use of AI tools	

12	Module 4: AI-Augmented Software Development	Reviewing and validating AI-generated code	Review: “Accept, Edit, or Reject?”	
13	Module 4: AI-Augmented Software Development	Prompt engineering and tool tuning	- Build a feature with human + AI collaboration - CS5: design and implement test units for the developed features.	20
14	Final Review	Presentations: Team AI-augmented projects	- Final reflection paper: “How I Will Evolve as an Engineer” - CS5: Final reflection paper: “research-style paper or industry whitepaper analysis on an AI-related SE problem”	10
15	Project Demos	Project Demos	Project Demos	
16	Pre Final Week	Pre Final Week	Pre Final Week	
17	Exam	Exam	Exam	20

Course Policies:

Academic Integrity and Plagiarism

The goal of this course is your learning, so the work you submit must be your own. Write in your own words and cite sources to avoid plagiarism. Group work, tutoring, campus resources, or online tools (including AI) are fine, but your submissions must show your understanding. If you're unsure about academic integrity, ask me. Serious violations can mean a zero on the assignment or even expulsion, so please know the expectations. Visit the [OU Integrity Office](#) for more information on what constitutes plagiarism.

Make-up Policy

Unless explicitly stated otherwise specified in writing, please ensure all assignments are submitted by the designated date in the Ticket instructions. In the event of a delay, a 10% deduction will be applied for each day beyond the specified deadline. This policy is in place to maintain fairness and consistency. It's worth noting that, as software engineering professionals, it's our responsibility to ensure timely submission, avoiding any delays that may result in fines for our workplace. Make-up exams will not normally be permitted, and they are only available when required by university policy.

Attendance Policy

This course follows a synchronous format, requiring your attendance at all scheduled class sessions and labs in person. Exceptions are made for illness, unforeseen caretaking duties. I do not take attendance in class, while presentations will be scheduled during the class meeting time, and they should be done in person.

Technology & Generative AI Policy

You'll occasionally use your devices in class for polls, quick research, or checking Canvas. Please be mindful: multitasking on devices distracts others. If you get an urgent notification: medical, childcare and/or eldercare, step out quietly to take the call. Furthermore, in recognizing the lasting impact of AI tools, I encourage their use to improve your skills on using them. However, given that AI tools are not fully matured, it is the responsibility of the student to evaluate the content generated and learn how to effectively work with AI tools to achieve optimal results. This approach reflects our commitment to adapting and utilizing emerging technologies responsibly in the learning environment. It is essential to note that any direct copy-pasting without reading, understanding, analyzing, and actively working to enhance your skills will be considered academic misconduct.

University Policies

Mental Health Support Services:

Support is available for any student experiencing mental health issues that are impacting their academic success. Students can either be seen at the University Counseling Center (UCC) located on the second floor of Goddard Health Center or receive 24/7/365 crisis support from a licensed mental health provider through [TimelyCare](#). To schedule an appointment or receive more information about mental health resources at OU please call the UCC at 405-325-2911 or visit [University Counseling Center](#). The UCC is located at 620 Elm Ave., Room 201, Norman, OK 73019.

Title IX Resources and Reporting Requirement

The University of Oklahoma faculty are committed to creating a safe learning environment for all members of our community, free from sex-based discrimination, including sexual harassment, domestic and dating violence, sexual assault, and stalking, in accordance with Title IX. There are resources available to those impacted, including: speaking with someone confidentially about your options, medical attention, counseling, reporting, academic support, and safety plans. If you have (or someone you know has) experienced any form of sex-based discrimination or violence and wish to speak with someone confidentially, please contact [OU Advocates](#) (available 24/7 at 405-615-0013) or [University Counseling Center](#) (M-F 8 a.m. to 5 p.m. at 405-325-2911). Because the University of Oklahoma is committed to the safety of you and other students, and because of our Title IX obligations, I, as well as other faculty, Graduate Assistants, and Teaching Assistants, are mandatory reporters. This means that we are obligated to report sex-based violence that has been disclosed to us to the Institutional Equity Office. This means that we are obligated to report sex-based violence that has been disclosed to us to the Institutional Equity Office. This includes disclosures that occur in: class discussion, writing assignments, discussion boards, emails and during Student/Office Hours. You may also choose to report directly to the Institutional Equity Office. After a report is filed, the Title IX Coordinator will reach out to provide resources, support, and information and the reported information will remain private. For more information regarding the University's Title IX Grievance procedures, reporting, or support measures, please visit [Institutional Equity Office](#) at 405-325-3546.

Reasonable Accommodation Policy

The University of Oklahoma (OU) is committed to the goal of achieving equal educational opportunity and full educational participation for students with disabilities. If you have already established reasonable accommodations with the Accessibility and Disability Resource Center (ADRC), please log into iAdvise to request your semester accommodations as soon as possible and contact me privately, so that we have adequate time to arrange your approved academic accommodations.

If you have not yet established services through ADRC, but have a documented disability and require accommodations, please complete [ADRC's pre-registration form](#) to begin the registration process. ADRC facilitates the interactive process that

establishes reasonable accommodations for students at OU. For more information on ADRC registration procedures, please review their [website](#). You may also contact them at (405)325-3852 or adrc@ou.edu, or visit www.ou.edu/adrc for more information.

Note: disabilities may include, but are not limited to, mental health, chronic health, physical, vision, hearing, learning and attention disabilities, pregnancy-related. ADRC can also support students experiencing temporary medical conditions.

Religious Observance

It is the policy of the University to excuse the absences of students that result from religious observances and to reschedule examinations and additional required classwork that may fall on religious holidays, without penalty.

[\[See Faculty Handbook 3.15.2\]](#)

Adjustments for Pregnancy and Related Issues

Should you need modifications or adjustments to your course requirements because of pregnancy or a pregnancy-related condition, please request modifications via the [Institutional Equity Office website](#) or call the Institutional Equity Office at 405/325-3546 as soon as possible. Also, see the Institutional Equity Office [FAQ on Pregnant and Parenting Students' Rights](#) for answers to commonly asked questions.

Final Exam Preparation Period

Pre-finals week will be defined as the seven calendar days before the first day of finals. Faculty may cover new course material throughout this week. For specific provisions of the policy please refer to OU's [Final Exam Preparation Period policy](#).

Emergency Protocol

During an emergency, there are official university [procedures](#) that will maximize your safety.

Severe Weather: If you receive an OU Alert to seek refuge or hear a tornado siren that signals severe weather.

1. Look for severe weather refuge location maps located inside most OU buildings near the entrances.
2. Seek refuge inside a building. Do not leave one building to seek shelter in another building that you deem safer. If outside, get into the nearest building.
3. Go to the building's severe weather refuge location. If you do not know where that is, go to the lowest level possible and seek refuge in an innermost room. Avoid outside doors and windows.
4. Get in, Get Down, Cover Up
5. Wait for official notice to resume normal activities.

Additional [Weather Safety Information](#) is available through the Department of Campus Safety.

The University of Oklahoma Active Threat Guidance

The University of Oklahoma embraces a Run, Hide, Fight strategy for active threats on campus. This strategy is well known, widely accepted, and proven to save lives. To receive emergency campus alerts, be sure to update your contact information and preferences in the account settings section at one.ou.edu.

RUN: Running away from the threat is usually the best option. If it is safe to run, run as far away from the threat as possible. Call 911 when you are in a safe location and let them know from which OU campus you're calling from and location of active threat.

HIDE: If running is not practical, the next best option is to hide. Lock and barricade all doors; turn off all lights; turn down your phone's volume; search for improvised weapons; hide behind solid objects and walls; and hide yourself completely and stay quiet. Remain in place until law enforcement arrives. Be patient and remain hidden.

FIGHT: If you are unable to run or hide, the last best option is to fight. Have one or more improvised weapons with you and be prepared to attack. Attack them when they are least expecting it and hit them where it hurts most: the face (specifically eyes, nose, and ears), the throat, the diaphragm (solar plexus), and the groin.

Please save OUPD's contact information in your phone.

NORMAN campus: *For non-emergencies call (405) 325-1717. For emergencies call (405) 325-1911 or dial 911.*

TULSA campus: *For non-emergencies call (918) 660-3900. For emergencies call (918) 660-3333 or dial 911.*

Fire Alarm/General Emergency:

If you receive an OU Alert that there is danger inside or near the building, or the fire alarm inside the building activates:

1. *LEAVE* the building. Do not use the elevators.
2. *KNOW* at least two building exits
3. *ASSIST* those that may need help
4. *PROCEED* to the emergency assembly area
5. *ONCE safely outside, NOTIFY first responders of anyone that may still be inside building due to mobility issues.*
6. *WAIT* for official notice before attempting to re-enter the building.

[OU Fire Safety on Campus](#)

Office of Access and Opportunity's Belonging Statement

Why You Belong at the University of Oklahoma: The University of Oklahoma fosters an inclusive culture of respect and civility, belonging, and access, which are essential to our collective pursuit of excellence and our determination to change lives. The unique talents, perspectives, and experiences of our community enrich the learning, and

working environment at OU, inspiring us to harness our innovation, creativity, and collaboration for the advancement of people everywhere.

Course Reflection Survey

You'll receive a Course Reflection Survey at the end of each semester for each course that you are enrolled in. I strongly encourage you to complete this survey. Your feedback can help me adjust my class for future semesters to help other students be successful. Your feedback is confidential and I will only receive it after final grades are due. Course Reflection Survey results may also factor into teaching evaluations and annual performance reviews and are shared with department and program chairs.

Copyright Statement, for in-person or online courses

Sessions of this course may be recorded or live-streamed. These recordings are the intellectual property of the individual faculty member and may not be shared or reproduced without the explicit, written consent of the faculty member. In addition, privacy rights of others such as students, guest lecturers, and providers of copyrighted material displayed in the recording may be of concern. Students may not share any course recordings with individuals not enrolled in the class or upload them to any other online environment.

Pre-Finals Week Policy

During pre-finals week, all normal class activities will continue; however, no assignment, test, or examination accounting for more than 3% of the course grade may be assigned, unless it is assigned in advance of pre-finals week and worth less than 10%, or scheduled at least 30 days prior if worth more than 10%. No activity or field trip may be scheduled that conflicts with another class. There are some exceptions and nuances, so please review the [Final Exam Policies](#) prior to designing your course schedule.